ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Experience 4 – Thermal camera

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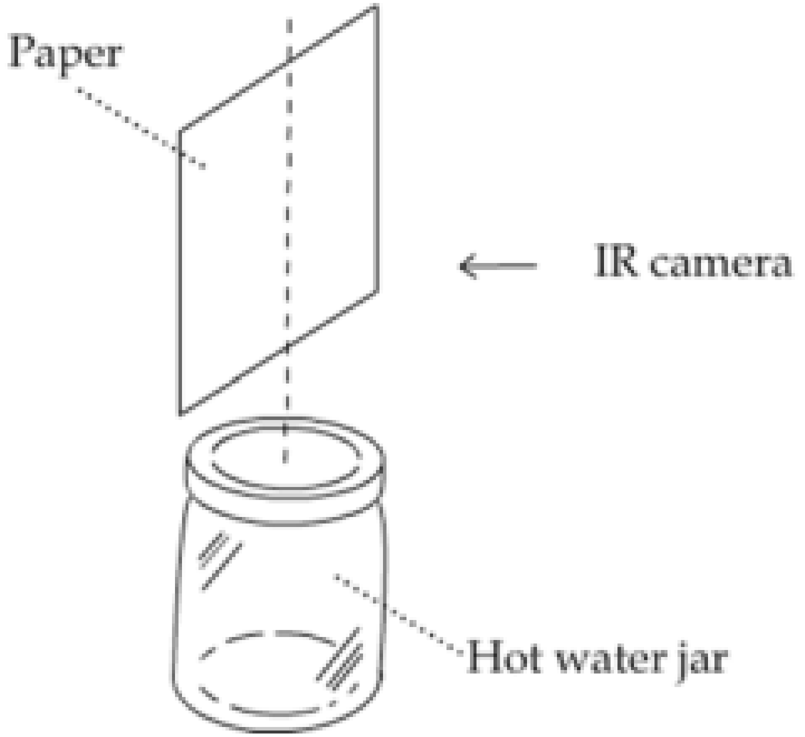
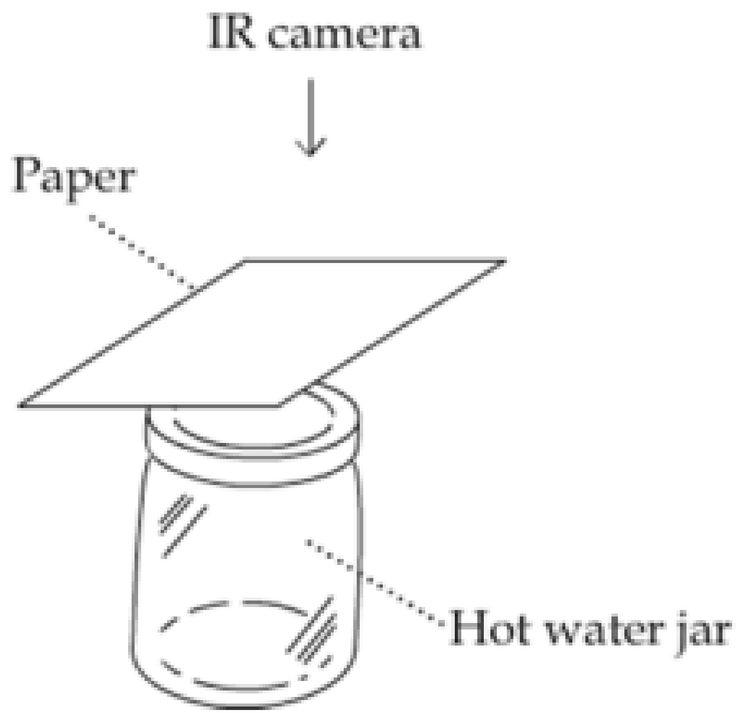
Part 1: natural convection

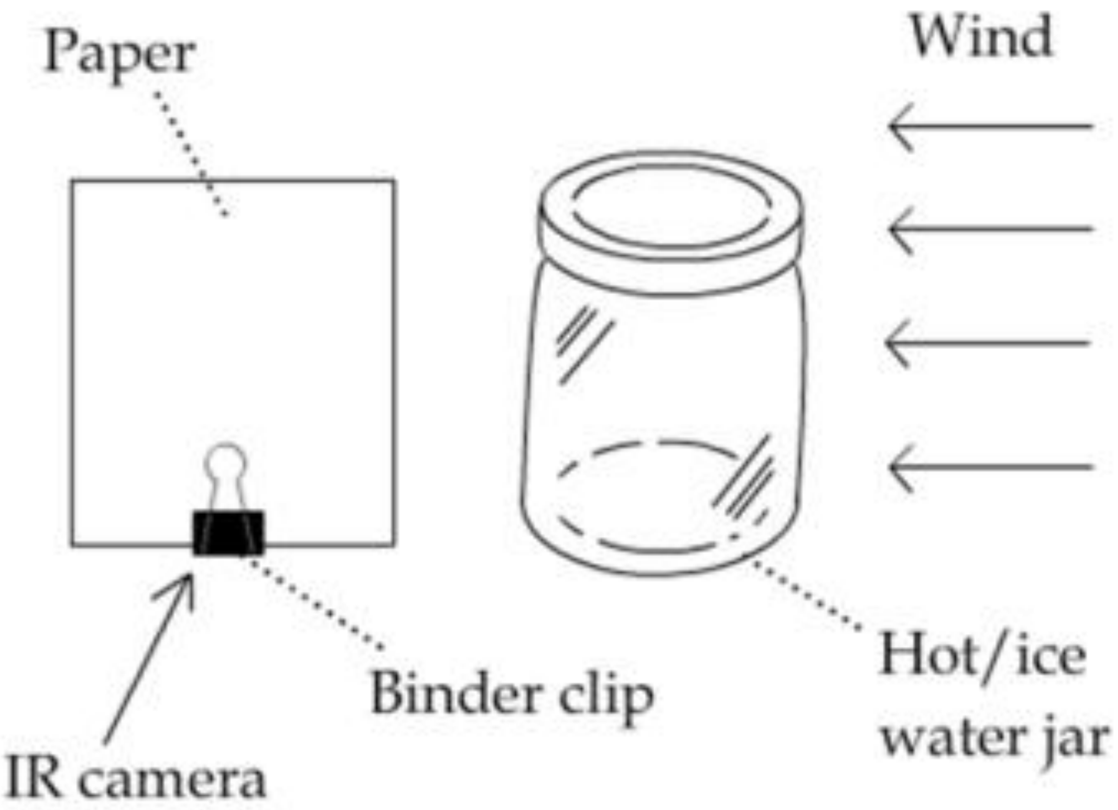
- An IR camera
- A few pieces of paper
- A glass of hot water (>80°C)





Part 2: Material and tools

- An IR camera
- A glass of hot water
- A hand-held fan



Main differences across groups

- Choice of cursor, box: (number), shape, position, and size
- Selection and position of the fan (distance, angle, height)
- Order of the two setups
- Relative position of the cup

Some questions

1. Did you observe anything visibly (with your eyes, with the camera in visible mode, …)?

2. Was the temperature pattern distributed as expected / predicted? If not, can you say why? Explain.

3. Was the diffusion homogeneous? Was the process continuous or intermittent?
4. How can you distinguish the effects of heat transfer on a piece of paper from a hot water jar through radiation and convection?
5. How much does the paper above warm?

6. What can you do to increase the heat transfer from the hot water jar to the paper through forced convection? List all possible methods and explain them

7. Can you find any similarities of atmospheric or daily life phenomena that you can represent with this experiment? Please explain, possibly for all phases (horizontal / vertical paper, with and without fan).

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Report

Instructions

- Maximum 7 pages
- Deadline 2 weeks from the experiment, but please allow enough time for evaluation if you aim to do examination on the 4 June
  - 4 June for exams after 4 June
  - 31 May/1 June for exams on 4 June
- Provide images or other supporting materials (e.g., tables with quantities) whenever necessary to support your statements.

For the report, you have received all the images collected during the experience in jpg (or any other format you prefer), along with a report statistics for the central position of the image and other positions (cursor 3x3, box, … explanations in the lab).

Eventually, csv matrix files can also be exported. Given the objectives of the experience, it will suffice to comment the selected images eventually with some statistics to discuss what you have observed during the experiment.

Eventually R and Python libraries to work with thermal images were also presented in the previous lecture.

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